

MA 301
Final
Spring - 2014

Problem	points	out of
1		4
2		4
3		6
4		14
5		6
6		6
7		9
8		15
9		6
10		20
Score		90

Name: _____

HR: _____

Signature: _____

READ BEFORE YOU START
* Show all work. * Change your calculator mode to RADIANS. * Sloppy solutions may be penalized. * Use calculators ONLY for computations. * You must write up every problem in a clear and organized way. * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Suppose the force field $\mathbf{F} = [2xy + z, x^2 + z^2, x + 2yz]$ acts upon a point mass along some curve C from $A : (1, 3, 0)$ to $B : (1, -1, 1)$.

There are two ways to compute the work done in moving the point mass from A to B . Give one of them.

Problem 2: Give the other method from **Problem 1**.

Problem 3: Let $\mathbf{F} = [2x^2y, -2xy^2]$ and C be given by $x^2 + y^2 = 5$ oriented counter-clockwise. Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ using green's theorem.

Problem 4: Let $z_1 = -2 + 2i$, $z_2 = -i$, $z_3 = 1 - i$. Answer the following in the (*specified form*):

(a) (*Standard Form*) $z_1 z_2 =$

(b) (*Standard Form*) $z_3 - z_1 z_2 =$

(c) $|z_1 z_2| =$

(d) Convert z_2 to Euler's Polar form:

(e) Convert z_3 to Euler's Polar form:

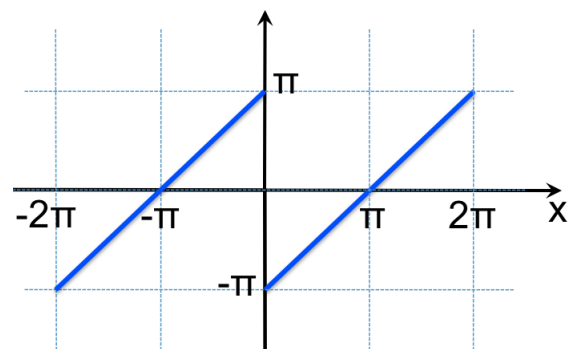
(f) (*Euler's Polar Form*) $(z_3)^{76} =$

(g) (*Euler's Polar Form*) Solve for x : $\frac{1}{\sqrt{2}}x^3 - z_3 = 0$ (explicitly write out all solutions)

Problem 5: Let $f(x) = \begin{cases} \pi & -\frac{\pi}{2} < x < 0 \\ -\pi & 0 < x < \frac{\pi}{2} \end{cases}$ and $f(x) = f(x + \pi)$

Find the Fourier Series of $f(x)$ in terms of: (i) $e^{i\pi nx/L}$ and (ii) $\cos\left(\frac{\pi}{L}nx\right)$ & $\sin\left(\frac{\pi}{L}nx\right)$

Problem 6: Completely Set-up (do not compute) the Fourier coefficients for the function shown below. Assume it periodically extends along the x -axis.



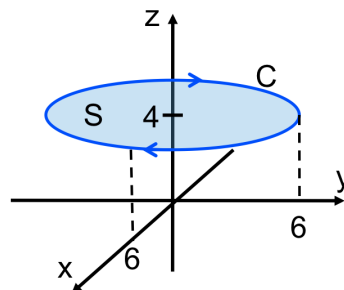
Problem 7: Answer the following true/false and short-answer questions:

T	F	The line integral of any vector field $\mathbf{F}$ around any closed curve C is zero
T	F	Let C be an ellipse. If $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$, then $\mathbf{F}$ is path-independent.
T	F	If $\mathbf{F}$ is path-independent, then there is a potential function for $\mathbf{F}$
T	F	If $\mathbf{F}$ is path-independent and C is any closed curve, then $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$
T	F	Suppose $\mathbf{F} = [1, 1, 1]$ and S is a closed surface with positive orientation. Then $\iint_S \mathbf{F} \cdot \mathbf{n} dA > 0$.
T	F	Let S be a sphere of radius 1 centered at the origin and oriented outward. If $\iint_S \mathbf{F} \cdot d\mathbf{S} = 0$, then $\mathbf{F} = [0, 0, 0]$
T	F	Let S be the surface of a cube centered at the origin with side length 2 and oriented outward. If $\mathbf{F} = 2\mathbf{i} - \mathbf{k}$ then the flux of $\mathbf{F}$ through S is 27.
T	F	If $\mathbf{F}$ is a vector-field such that $\text{div}\mathbf{F} = 3$ and S is a closed surface oriented outward, then $\iint_S \mathbf{F} \cdot d\mathbf{S}$ is equal to one-third the volume enclosed by S .
T	F	Suppose $f(x) = f(x + 6)$ for all x , then we say f is periodic with period 12.

Problem 8: Answer the following short-answer questions:

(a) Let $\mathbf{F}(x, y, z) = [zy - 3y, xz - z, xy + 1]$

$$\int_C \mathbf{F} \cdot d\mathbf{r} =$$



(b) Let $\mathbf{F}$ vector field such that $\text{curl } \mathbf{F} = \mathbf{0}$. Match each expression in the left column with the most appropriate term in the right column.

$\int_C \mathbf{F} \cdot d\mathbf{r}$	conservative
$\oint_D \mathbf{F} \cdot d\mathbf{r}$	path independent
$\mathbf{F}$	zero

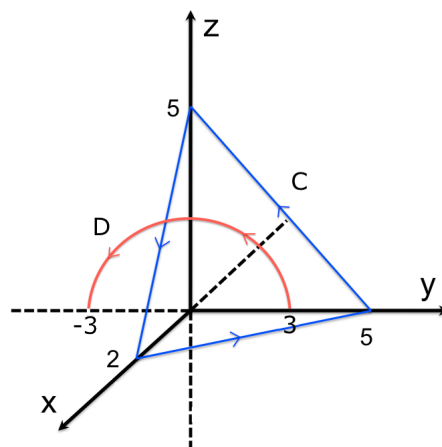
(c) Match each the Theorem in the left column with the most appropriate statement in the right column.

Divergence Theorem	The surface of a full sphere of radius 4
Stokes' Theorem	The surface of a half-sphere of radius 4
Green's Theorem	A disk of radius 4

(d) Let $\mathbf{F} = \nabla \left(\frac{y-1}{y+z+1} \right)$ and let C and D be shown in the figure. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} =$$

$$\int_D \mathbf{F} \cdot d\mathbf{r} =$$



(e) Let $\mathbf{F} = [6x, ye^z, -e^z]$ and S be the surface (oriented outward) of the box $|x| \leq 1, |y| \leq 2, 0 \leq z \leq 1$.

$$\iint_S \mathbf{F} \cdot \mathbf{n} dA =$$

Problem 9: Set-up the integrals (to the point you would use an integration technique) needed to compute:

(a) The mass of the wire with shape given by $\mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} + \frac{2}{3}t^{3/2}\mathbf{k}$, $0 \leq t \leq 2$ & density $\delta(x, y, z) = 3\sqrt{5+x}$.

(b) The surface area of S where S is given by the parameterization: $\mathbf{r}(u, v) = [u, v, u^3]$, $0 \leq u \leq 1$, $-2 \leq v \leq 2$

Problem 10: Answer all of the following questions based on the figure below.

$$\mathbf{F}(x, y, z) = [x + y, y^2, 4 - z]$$

S_B : Part of the $z = 0$ plane with $-2 \leq x \leq 2, 0 \leq y \leq 3$

S_L : Part of the $y = 0$ plane with $0 \leq z \leq 4 - x^2, -2 \leq x \leq 2$

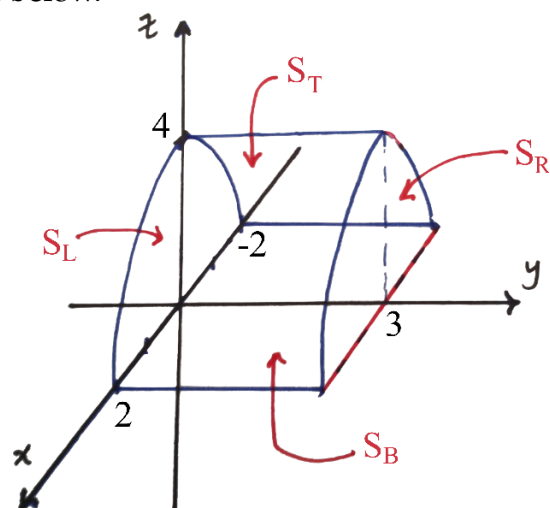
S_R : Part of the $y = 3$ plane with $0 \leq z \leq 4 - x^2, -2 \leq x \leq 2$

S_T : Part of the parabolic cylinder $z = 4 - x^2$ with $0 \leq y \leq 3, -2 \leq x \leq 2$

S : Total closed surface in the figure (i.e. $S = S_B \cup S_L \cup S_R \cup S_T$)

Hint: $\text{Area}(S_L) = \text{Area}(S_R) = 2 \int_0^2 (4 - x^2) dx$

(a) Compute the flux through S_B directly.



(Assume outward orientation)

(b) Compute the flux through S_L directly.

(c) Compute the flux through S_R directly.

(d) Compute the flux through S_T directly.

(e) Compute the flux through S using (a)-(d).

(f) Compute the flux through S using the Divergence Theorem.

(g) If C_T is the boundary of S_T (*oriented counter-clockwise*), compute $\int_{C_T} \mathbf{F} \cdot d\mathbf{r}$ using Stokes' Thm